



Which video game genre(category) is trending throughout the year ?

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PROBLEM STATEMENT

- The video game industry is the most trending industry globally which makes impact on growth of GDP.
- Video game companies need to identify which video game category is trending globally.
- Thus, finding best algorithm for prediction becomes crucial task for Data Scientist.

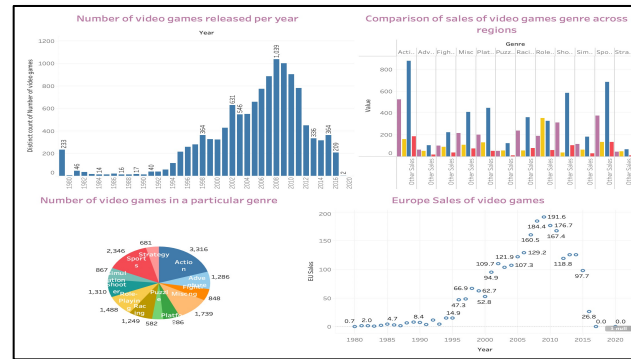
INTRODUCTION

- Recent studies shows that 1/3rd of global population widely play video games to reduce stress. A record was registered by a person named Carrie by playing nonstop games for 138 hours. “*Minecraft*” is the most played game which sold 33 billion units till date.
- In the year 2020, USA generated \$155 billion revenue by selling video games worldwide.
- The goals of this project is to preprocess data, visualize and apply appropriate models on dataset. The project will deliver a result that will address above research problem.
- The Dataset consist of video game list which is based on ranking of games from year 1980-2020, the platform it was released and sales of different regions.

DATA PREPARATION

- Analyzed the data and found that there were few null values. Dropped all the null values.
- Once the null values were dropped then data was changed to 16291 rows and 11 columns.
- The data type of Global Sales was changed from object to float and then it was used to calculate predictions.
- Further explored the data then distributed the dataset into numerical and categorical features.

DATA VISUALIZATION – TABLEAU DASHBOARD



EXPERIMENTATION & METHODOLOGY

1. Classification Algorithm

- Used Classification algorithm(KNN) to solve research problem.
- Studies from authors shows linear regression algorithm provides accurate results.

2. Cross Validation

- Used cross validation score as a parameter to find the best k.
- Cross validation score is also used to find max and min error.
- The min. error k will be the best k.

3. Comparing algorithms,

- Compared within the algorithm using brute force algorithm and K-d tree to find the best model.
- Compared odd and even numbers of K. For odd numbers used Brute force algorithm as it works best for odd numbers and for even numbers used K-d tree.
- Compared classification report, confusion matrix and plot heatmap to find which game genre was accurately predicted by model.

RESULTS

1. Operations to find best K

- Min Error Value in odd and even numbers

Error in range (0,50)		
	Min	Max
ODD	3	49
EVEN	2	48

Actual Error Value		
ODD	3	0.01
	49	0.03
EVEN	2	0.01
	48	0.03

- In odd numbers we obtain k= 3 is best as its error value is 0.0157968

2. Comparing Brute-force and K-d tree with KNN algorithm

- We found out that the prediction was not same but quite similar. We concluded that K-d tree is the best model for our dataset as it provides a little higher accuracy rate.

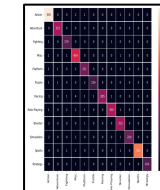
Algorithm	Value of K	Accuracy Rate Predicted
Brute -force	ODD K= 3	98.64%
K-d tree	EVEN K = 2	98.89%

3. Compared classification report with confusion matrix to label the predicted values.

- We found by comparing both that in action genre total 804 games but out of which 800 are correctly classified and 4 are wrongly classified. Similarly, in Fighting genre there are in total 181 out of which 2 are wrongly classified and in adventure genre all games are correctly classified.

Actual Values Types for K-Nearest Neighbors Classification										
Actual \ Predicted	Action	Adventure	Fighting	Platform	Puzzle	Racing	Role-Playing	Shooter	Simulation	Sports
Action	804	0	0	0	0	0	0	0	0	0
Adventure	0	804	0	0	0	0	0	0	0	0
Fighting	0	0	181	0	0	0	0	0	0	0
Platform	0	0	0	181	0	0	0	0	0	0
Puzzle	0	0	0	0	181	0	0	0	0	0
Racing	0	0	0	0	0	181	0	0	0	0
Role-Playing	0	0	0	0	0	0	181	0	0	0
Shooter	0	0	0	0	0	0	0	181	0	0
Simulation	0	0	0	0	0	0	0	0	181	0
Sports	0	0	0	0	0	0	0	0	0	181

Classification Report using K-Nearest Neighbors				
Genre	precision	recall	f1-score	support
Action	0.99	1.00	0.99	804
Adventure	1.00	1.00	1.00	333
Fighting	0.99	0.99	0.99	181
Platform	0.99	1.00	0.99	427
Puzzle	0.99	0.99	0.99	181
Racing	0.99	0.99	0.99	181
Role-Playing	0.99	0.99	0.99	181
Shooter	0.99	0.99	0.99	181
Simulation	1.00	0.99	0.99	224
Sports	1.00	0.99	0.99	181
Strategy	1.00	0.99	0.99	181
accuracy	0.99			
macro avg	0.99	0.99	0.99	4873
weighted avg	0.99	0.99	0.99	4873



WHY KNN?

- K-nearest neighbor** is simple and basic classification algorithm.
- Instead of comparing machine learning algorithm which is done by every author in research.
- Tested different ways in KNN algorithm for prediction on dataset.

CONCLUSIONS

- This project compares the performance of different models on the video game dataset. In this we have compared within the algorithm to predict which is the better.
- As a result, in odd number, we obtain best K as 3 and in even number we obtain 2 as best K
- The result for Brute-force and k-d tree was not not same but similar accuracy rate was predicted.
- At last concluded that K-d tree is better than Brute-force.
- We obtain highest accuracy rate that is 98.89%.
- In future work we can compare heatmaps, by performing train and test on same dataset randomly.